Code No: 113BS **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, May/June - 2019
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is Diminished Radix Complement? [2]
b) What is Minterms and Maxterms? [3]
c) What is Prime Implicants? [2]
d) What is don't-care conditions? Explain it uses. [3]
e) Explain about 4-bit Binary Adder. [2]
f) Explain about Multiplexers with three-state gates. [3]
g) Explain about characterstic equation for jk flip flop. [2]
h) Define the following terms of a flip flop.
i) Hold time ii) Set up time iii) Propagation delay time. [3]
i) Explain about sequential memory. [2]
j) Explain about the memory read and write operations. [3]

PART-B

(50 Marks)

2.a) If the numbers $(+9,742)_{10}$ and $(+641)_{10}$ are in signed magnitude format, their sum is $(+10,383)_{10}$ and requires five digits and a sign. Convert the numbers to signed-10's complement form and find the following sums:
(i) (+9,742) + (+641) (ii) (+9,742) + (-641)
b) Convert the hexadecimal number 64CD H to binary and then convert it from binary to octal. [5+5]

OR

3.a) Express the complement of the following function F in sum of minterms form.
F(A,B ,C, D) = ∑(2, 4, 7, 10, 12, 14)
b) Show that the dual of the exclusive OR is equal to its complement. [5+5]

4.a) Simplify the following Boolean expressions, using four-variable maps:
i) A′B′C′D′ + AC′D + B′CD′ + A′BCD + BC′D
ii) A′B′C′D′ + BC′D + A′C′D + A′BCD + ACD′
b) Find all the prime implicants for the following Boolean functions, and determine which are essential: F (w, x, y, z) = ∑ (0, 2, 4, 5, 6, 7, 8, 10, 13, 15) [5+5]

OR

5.a) Draw a NAND logic diagram that implements the complement of the following function:
F(A,B,C,D) = ∑(0, 1, 2, 3, 6, 10, 11, 14).
b) Find the minterms of the following Boolean expressions by first plotting each function in a map:
(i) xy + yz + xy'z (ii) C'D + ABC' + ABD' + A'B'D. [5+5]

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6. Design a combinational circuit with three inputs, x, y and z and three outputs, A, B and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is two less than the input. [10]

OR

7.a) Implement a full adder with two 4 × 1 multiplexers.

b) Design a four-input priority encoder with input D_0 having the highest priority and input D_3 the lowest priority. [5+5]

8.a) Draw the logic diagram of a four-bit binary-ripple countdown counter using flip-flops that trigger on the positive edge of the clock.

b) Realize D-latch using R-S latch. How it is different from D-flip flop. Draw the circuit using NAND gates and explain. [5+5]

OR

9.a) Draw and explain with the help of truth table the logic diagram of a master slave D flip-flop using NAND gates. With active low preset and clear and with negative edge triggered clock.

b) Compare merits and demerits of ripple and synchronous counters. [5+5]

10. A 16K×4 memory uses coincident decoding by splitting the internal decoder into X-selection and Y-selection.

a) What is the size of each decoder, and how many AND gates are required for decoding the address?

b) Determine the X and Y selection lines that are enabled when the input address is the binary equivalent of 6,000. [10]

OR

11.a) Explain about different types of ROM.

b) Explain about Cache memory. [5+5]

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